
Container Terminal Operations Analytics

Enterprise Power BI Solution

Technical Documentation and Implementation Book

Data source: Terminal Operating System export
Scope: 98,052 operational records, 33 source fields, calling years
2025–2026
Platform: Microsoft Power BI Desktop and Power Query

Prepared by Container Terminal Business Intelligence Team
For enterprise handover, academic evaluation, maintenance, and audit

Version 1.0 – August 2026

Copyright and Document Control

This book documents a Power BI analytical solution developed for container terminal operations. It is intended for authorized project stakeholders. Project-specific operational data, terminal identities, commercial arrangements, and sensitive container details must be governed according to the terminal’s data-classification and access-control policies.

Document title	Container Terminal Operations Analytics: Enterprise Power BI Solution
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Source authority	User-provided project decisions and implementation history

Project Team

This page identifies the project delivery team and provides a controlled location for future updates to names, roles, organizational affiliations, and approval responsibilities. Replace the bracketed placeholders in `main.tex` with the approved project information.

Team Identification	
Team name	Container Terminal Business Intelligence Team
Organization / terminal	[Organization / Terminal Name]
Project supervisor	[Project Supervisor / Sponsor]

Table 1: Project team members and responsibilities

No.	Team member	Role and responsibility
1	[Team Member Name]	[Role / Responsibility]
2	[Team Member Name]	[Role / Responsibility]
3	[Team Member Name]	[Role / Responsibility]
4	[Team Member Name]	[Role / Responsibility]

Project Placeholder
Add or remove team-member rows as needed. Recommended role descriptions include project lead, Power BI solution architect, Power Query / ETL developer, semantic-model developer, DAX and KPI specialist, container-terminal subject-matter expert, data-quality reviewer, dashboard designer, project sponsor, and technical approver.

Document Approval

Prepared by	[Name and role]	Date:	[DD Month YYYY]
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Approved by	[Name and role]	Date:	[DD Month YYYY]

Dedication

To the operations, planning, engineering, safety, and information-technology professionals whose daily work keeps container terminals moving safely and efficiently.

Preface

This publication transforms the complete implementation history of a first enterprise Power BI project into a maintainable technical book. It records the reasoning behind every significant modeling decision, including why a flat Terminal Operating System export was converted to a star schema, how technical keys and dimensions were created, how active and inactive date relationships were applied, why repeated quay-crane working hours were isolated into a separate fact, and how validation pages were used before executive dashboards were built.

The book is intentionally chronological. It begins with the original source and business process, progresses through Power Query and semantic modeling, and concludes with operational dashboards, governance, optimization, and future enhancements. Whenever an exact project detail was not available, a clearly marked placeholder is included rather than an invented value.

Acknowledgments

The architecture follows established dimensional-modeling principles and Microsoft Power BI guidance. The business framing reflects standard container-terminal operating concepts while remaining constrained to the supplied source fields and validated project decisions.

Abstract

The project converts a 98,052-row, 33-column export from a Terminal Operating System into an enterprise Power BI semantic model for container terminal operations. The solution uses Power Query to stage, clean, normalize, deduplicate, and enrich the source; generates dimension keys and a central container-operations fact; establishes controlled many-to-one relationships; creates an explicit date dimension with role-playing entry and exit relationships; validates technical keys and data completeness; maps container length to TEU; isolates repeated quay-crane working-hour values into a second fact; and publishes governed DAX measures for flow, inventory, dwell, TEU, weight, vessel calls, quality, and QC productivity. Report pages include model, TEU, weight, and QC validation, Container Flow and Dwell Performance, and an Executive Overview with prior-period indicators. The documentation preserves final formulas, known limitations, rejected approaches, and future requirements such as historical inventory snapshots, berth and gate event data, detailed yard locations, and financial facts.

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List of Acronyms

BI	Business Intelligence
DAX	Data Analysis Expressions
DG	Dangerous Goods
ETL	Extract, Transform, Load
FCL	Full Container Load
IMDG	International Maritime Dangerous Goods
KPI	Key Performance Indicator
LCL	Less than Container Load
M	Power Query formula language
POL	Port of Loading
POD	Port of Discharge
QC	Quay Crane
RLS	Row-Level Security
TEU	Twenty-foot Equivalent Unit
TOS	Terminal Operating System

Project Introduction and Scope

1.1 Project objective

The objective was to build an enterprise-grade analytical solution for container terminal operations using Microsoft Power BI. The project began from a single Excel export rather than a prepared data warehouse. The implementation therefore included business interpretation, data profiling, Power Query transformation, dimensional modeling, DAX measure engineering, quality assurance, and dashboard construction.

The solution was designed to answer operational and management questions while preserving clear boundaries around what the source can and cannot support. The available source supports container flow, vessel and voyage workload, operator volume, cargo mix, container specification, port connectivity, weight, damage condition, dangerous-goods source values, dwell, and equipment association. It does not independently provide complete berth intervals, yard-slot geometry, truck queue milestones, rail-call events, detailed inspection history, financial transactions, or historical inventory snapshots.

1.2 Source summary

The validated fact contains **98,052 rows**. The source contains 33 columns and covers vessel calling years 2025 and 2026. Each row was treated provisionally as a container operational record associated with a vessel call and an entry/exit lifecycle. A generated `ContainerOperationKey` guarantees technical row uniqueness; it does not by itself prove that each row is a unique physical terminal visit.

Figure Placeholder

Original TOS Excel export showing the 33 source columns and representative records

Figure 1.1: Original TOS Excel export showing the 33 source columns and representative records

1.3 Project objectives

1. Preserve the source as a traceable staging layer.
2. Convert repeated descriptive attributes into reusable dimensions.
3. Create a fact table with explicit foreign keys and numeric measures.
4. Establish a governed date model for entry and exit analysis.
5. Prevent invalid aggregation of repeated QC working-hour values.
6. Publish validated units, TEU, dwell, inventory, vessel, weight, quality, and QC measures.
7. Build validation pages before management dashboards.
8. Produce an auditable technical baseline suitable for future Fabric or warehouse migration.

1.4 Methodological principles

The design applies a star schema, with dimensions used for filtering and grouping and facts used for numeric aggregation, consistent with Microsoft guidance (Microsoft 2024). Power Query performs repeatable source shaping, while DAX performs filter-context-sensitive calculations. Relationships default to many-to-one and single-direction filtering. Unknown or unsupported values are preserved and flagged instead of silently deleted or forced into assumptions.

Container Terminal Business Context

2.1 End-to-end container lifecycle

A container terminal coordinates vessel, quay, yard, gate, rail, documentation, safety, and customer processes. The supplied dataset captures a compressed lifecycle through container identity, vessel call, entry and exit dates, transport classifications, cargo, dangerous-goods source values, damage condition, equipment association, stack days, and weight.

Container arrival → quay handling → yard storage → optional inspection/transfer → departure by vessel, truck, or rail

2.2 Vessel and quay operations

Vessel operations cover call identification, voyage assignment, vessel code and name, and the workload associated with a call. Quay cranes transfer containers between vessel and terminal. The source associates an equipment number and QC working hours with container rows, but the QC hours were found to be repeated at a higher grain. This led to a dedicated QC validation and performance fact rather than summing the source column.

2.3 Yard and dwell operations

The source provides `In Date`, `Out Date`, and `Stack Days`, supporting completed dwell and current open-record age. It does not contain block, bay, row, tier, or yard capacity; therefore, spatial yard occupancy and heat maps remain future enhancements.

2.4 Import, export, and transshipment

The transport and mode fields can segment inbound and outbound patterns, but their exact operational codes remain source-controlled. A combined transport-profile dimension was created to preserve status, delivery, dispatch, full/empty, LCL/FCL, mode, S/A, and inbound/outbound transport classifications without inventing semantics.

2.5 Management questions supported

The model supports questions such as:

- How many records and TEU entered or exited by date?
- How many open records and unique containers remain inside the latest extract?
- What are average, median, P90, and maximum completed dwell?
- Which operators, vessels, voyages, cargo types, commodities, and port pairs dominate workload?
- Which equipment groups have validated QC working hours, and what records or TEU are handled per hour?
- Which records have missing weight, unknown size, unmapped TEU, or inconsistent QC hours?

Source Data Assessment

3.1 Original fields

The source fields were:

Field	Initial business interpretation
Size Type	Combined equipment size/type or ISO-style code
Cntr Len	Nominal container length
Container No.	Physical container identifier
OPR	Operator or carrier code; exact commercial role requires confirmation
Status	Operational status code
Out Date	Exit or lifecycle completion date/time
Calling Seq.	Vessel-call sequence
Calling Year	Vessel-call year, 2025 or 2026
Cargo Type	Broad cargo classification
Cntr H&W	Height-and-width or dimensional classification
Cntr Type	Equipment type
Commodity	Commodity description
Damage Condition	Recorded container damage condition
Delivery	Delivery classification or status
EQU No.	Equipment identifier
QC Working Hours	Source QC-hours value, later validated at call/equipment grain
Dispatch Mode (In)	Inbound dispatch mode
FE	Full/empty status
IMDG	Dangerous-goods source value
In Date	Entry date/time
LCL FCL	LCL/FCL classification
Mode	Source operational mode
POD	Port of discharge code
POL	Port of loading code
S/A	Unresolved source abbreviation
Stack Days	Source-calculated stack/dwell days
Trans Type (In)	Inbound transport type
Trans Type (Out)	Outbound transport type
User Voyage	Operational voyage reference
Vessel Code	Vessel code
Vessel Name	Vessel name
Weight Group	Source-defined weight band
WGT	Source weight value

3.2 Grain assessment

The project did not assume that container number alone was unique. The fact uses `ContainerOperationKey`, a Power Query index, to identify each loaded row. Business duplicate analysis remains distinct from technical-key uniqueness. The final validation confirmed:

Source Records	98,052
Unique Operation Keys	98,052
Duplicate Operation Rows	0
Blank Operation Keys	0

Important Limitation

An indexed technical key makes each row unique even if two source rows are otherwise identical. It is not a substitute for a business duplicate test using container, vessel call, dates, and equipment.

3.3 Initial data risks

The assessment identified conditional blanks, inconsistent text casing and spaces, unknown size values, possible free-text commodity variation, repeated QC working hours, date-role ambiguity, missing weight, and source codes without an external business dictionary. These risks were addressed through cleaning, mapping, match-status flags, and validation pages rather than aggressive deletion.

4

Architecture and Implementation Methodology

4.1 Layered architecture

1. Excel TOS export and manually maintained reference tables.
2. Power Query staging query with source-preserving transformations.
3. Dimension and fact queries created by Reference.
4. Certified semantic model with star-schema relationships.
5. Central `_Measures` table with display folders.
6. Validation pages followed by operational and executive dashboards.

Figure Placeholder

End-to-end implementation workflow from Excel source through Power Query, semantic model, validation, and dashboards

Figure 4.1: End-to-end implementation workflow from Excel source through Power Query, semantic model, validation, and dashboards

4.2 Why Power Query rather than manually split Excel sheets

The dimensions were created in Power Query so that new source values can be captured during refresh and transformation logic remains reproducible. Excel sheets were retained only for manually governed

reference data such as the port master or future mapping tables. Manual splitting would introduce key maintenance, duplicate, and refresh risks.

4.3 Storage and refresh

The 98,052-row source is modest for Power BI Import mode. Full refresh is appropriate for the current scale. Incremental refresh is a future option when history, refresh frequency, and source query-folding justify it.

4.4 Naming conventions

Technical query and table names follow prefixes such as `Stg`, `Dim`, and `Fact`. Columns use compact PascalCase names. Measures are stored in `_Measures` and grouped into folders such as `01 Volume`, `02 Flow`, `03 Inventory`, `04 Dwell`, `07 Data Quality`, `08 Equipment and QC`, and `90 Technical`.

Power Query Staging and Preprocessing

5.1 Excel preparation and import

The source range was converted to a named Excel table, imported with *Transform Data*, and renamed `StgContainerOperations`. The staging query retained all source rows and became the common parent for fact and dimension references.

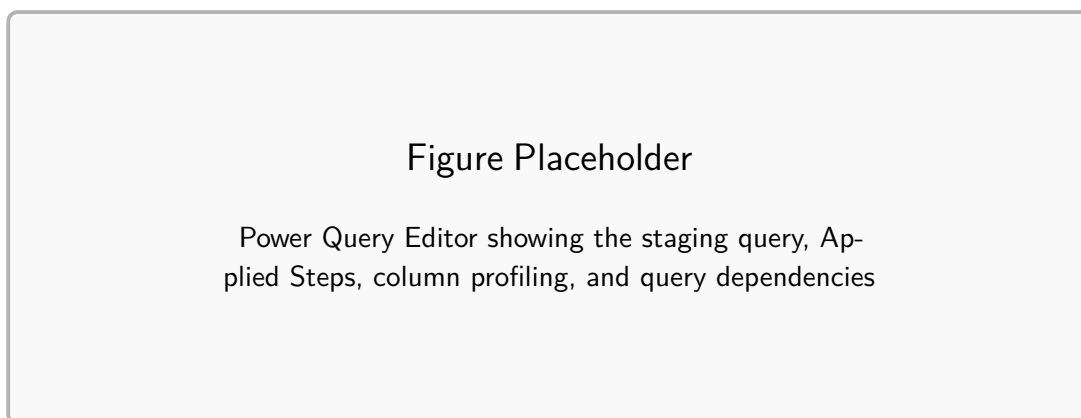


Figure 5.1: Power Query Editor showing the staging query, Applied Steps, column profiling, and query dependencies

5.2 Header standardization

Source headers were renamed for maintainability, for example:

Source	Standardized
Container No.	ContainerNumber
Calling Seq.	CallingSequence
Calling Year	CallingYear
Cntr Len	ContainerLength
Cntr H&W	ContainerHeightWidth
EQU No.	EquipmentNumber

QC Working Hours	QCWorkingHoursSource
In Date	InDate
Out Date	OutDate
Stack Days	StackDays
WGT	WeightValue

5.3 Data types and text cleaning

Dates were assigned Date or Date/Time types according to source precision. Weight, stack days, QC hours, and derived dwell were numeric. Operational codes were stored as text to preserve leading zeros. Text fields were trimmed and cleaned; selected codes were uppercased. Completely blank rows were removed, but conditional blanks were preserved.

5.4 Generated staging columns

5.4.1 SourceRowKey and UnitCount

An index starting at one created `SourceRowKey`, later renamed `ContainerOperationKey` in the fact. `UnitCount` was assigned 1 to every source record.

5.4.2 CompletedDwellDays

Listing 5.1: Power Query formula for completed dwell

```
if [InDate] = null or [OutDate] = null then
    null
else if [OutDate] < [InDate] then
    null
else
    Duration.TotalDays([OutDate] - [InDate])
```

Incomplete and invalid sequences remain blank rather than zero.

5.4.3 DateSequenceStatus

```
if [InDate] = null then
    "Missing In Date"
else if [OutDate] = null then
    "Active or Missing Out Date"
else if [OutDate] < [InDate] then
    "Invalid: Out Before In"
else
    "Valid Completed Record"
```

5.5 Load behavior

`StgContainerOperations` has Enable Load disabled. It remains available to child queries but is not duplicated in the semantic model.

Dimension Table Construction

6.1 General pattern

Every dimension followed the same controlled process:

1. Create a Reference from `StgContainerOperations`.
2. Keep only the dimension attributes.
3. Assign text types and apply the same cleaning logic as the fact.
4. Standardize blanks to `UNKNOWN` where appropriate.
5. Remove duplicate attribute combinations.
6. Add an index key starting at one.
7. Merge the dimension into the fact with a left outer join.
8. Expand only the key, add a match-status flag, and verify that the fact row count does not increase.
9. Remove repeated descriptive columns from the fact after validation.

Power Query left outer joins preserve all fact rows while adding matching dimension values (Microsoft 2025b).

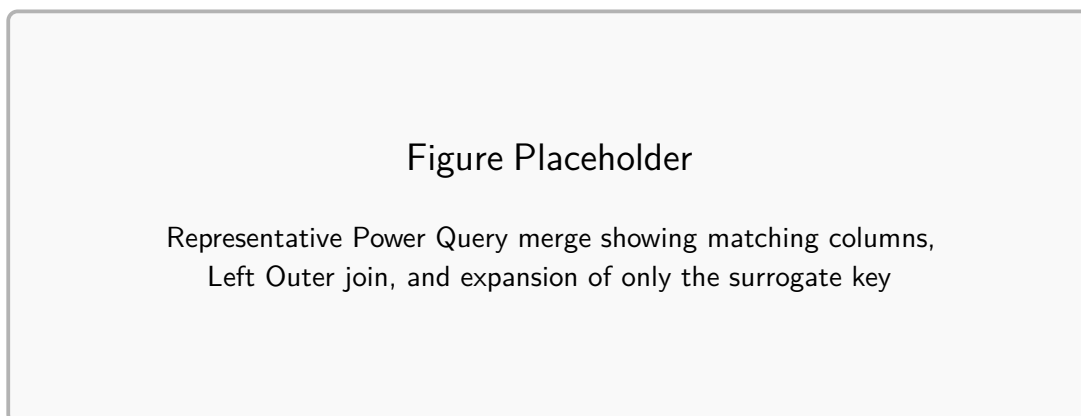


Figure 6.1: Representative Power Query merge showing matching columns, Left Outer join, and expansion of only the surrogate key

6.2 DimVesselCall

Attributes: CallingYear, CallingSequence, VesselCode, VesselName, UserVoyage.

Key: VesselCallKey .

The vessel-call dimension was created first because it provides the central maritime context. A readable VesselCallBusinessID concatenated year, sequence, vessel, and voyage. The unique combinations were indexed, then merged back into the fact using a left outer join.

6.3 DimOperator

Attributes: OperatorCode.

Key: OperatorKey .

The OPR source code was cleaned, deduplicated, indexed, and used for operator-level volume and dwell analysis. The exact operator role remains subject to business confirmation.

6.4 DimContainerSpecification

Attributes: SizeType, ContainerLength, ContainerHeightWidth, ContainerType.

Key: ContainerSpecificationKey .

This dimension centralizes equipment specifications and later stores TEUFactor and TEUMappingStatus.

6.5 DimCargo

Attributes: CargoType, Commodity.

Key: CargoKey .

Null or blank cargo descriptors were standardized to UNKNOWN. The source commodity was preserved rather than aggressively merged without an approved commodity master.

6.6 DimEquipment

Attributes: EquipmentNumber.

Key: EquipmentKey .

Equipment codes were cleaned and uppercased. QCWorkingHoursSource was explicitly excluded because it is a measurement, not a descriptive equipment attribute.

6.7 DimPortPOL

Attributes: POLCode plus PortMaster attributes.

Key: POLKey .

The role-playing port-of-loading dimension was enriched from the manually maintained port master using PortCode.

6.8 DimPortPOD

Attributes: PODCode plus PortMaster attributes.

Key: `PODKey` .

The separate port-of-discharge dimension permits simultaneous POL and POD filtering. Direct relationships between POL, POD, and the master were removed.

6.9 DimTransportProfile

Attributes: Status, Delivery, DispatchModeIn, FullEmptyStatus, LCLFCL, Mode, SA, Transport-TypeIn, TransportTypeOut.

Key: `TransportProfileKey` .

A junk-dimension design consolidates low-cardinality operational classifications. The distinct combination count must remain materially smaller than the fact.

6.10 DimDamageCondition

Attributes: DamageCondition.

Key: `DamageConditionKey` .

Source conditions were preserved, with UNKNOWN for blanks. An IsDamaged mapping was deferred until the business meaning of each value is approved.

6.11 Port master handling

The manually maintained PortMaster query contains PortCode, PortName, Country, Region, Latitude, and Longitude. It was merged into both port dimensions. Its Enable Load property was disabled so that it does not create redundant model tables or accidental one-to-one relationships.

6.12 IMDG and weight group decisions

A separate IMDG dimension was considered but not implemented in the final first model. The source IMDG field remains in the fact because its code semantics and possible multi-valued structure require further clarification. WeightGroup also remains in the fact until approved ranges and sort order justify a governed dimension.

Fact Table Construction

7.1 FactContainerOperations

`FactContainerOperations` was created as a Reference from the staging query. It retained the 98,052 operational rows and received the dimension foreign keys through sequential left outer merges. The final technical key is `ContainerOperationKey`.

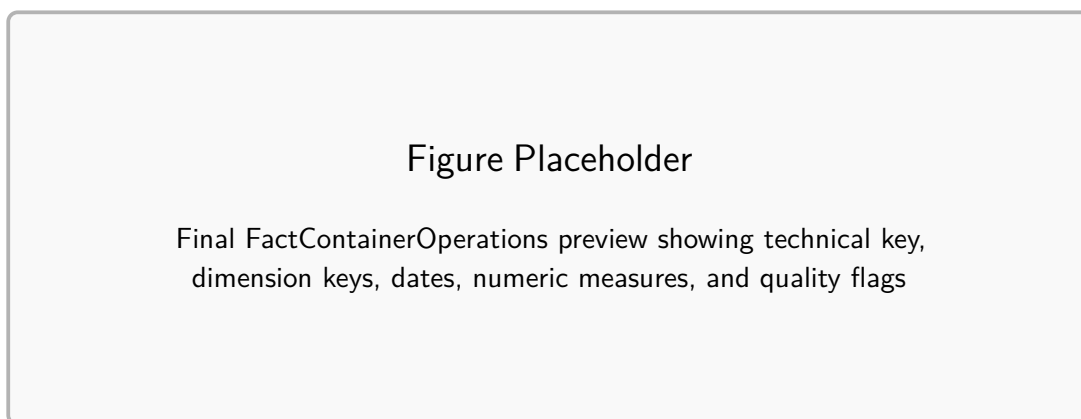


Figure 7.1: Final FactContainerOperations preview showing technical key, dimension keys, dates, numeric measures, and quality flags

7.2 Final fact structure

The first-model fact contains:

- Technical and degenerate identifiers: `ContainerOperationKey`, `ContainerNumber`.
- Foreign keys: `VesselCallKey`, `OperatorKey`, `ContainerSpecificationKey`, `CargoKey`, `EquipmentKey`, `POLKey`, `PODKey`, `TransportProfileKey`, `DamageConditionKey`.
- Direct attributes retained by design: `IMDG`, `WeightGroup`.
- Dates: `InDate`, `OutDate`, `InDateOnly`, `OutDateOnly`.
- Numeric data: `StackDays`, `WeightValue`, `QCWorkingHoursSource`, `UnitCount`, `CompletedDwellDays`.

- Current-state and quality data: IsCurrentlyInside, ActiveAgeDays, ActiveAgeBucket, and match-status columns.

7.3 Fact-level custom columns

7.3.1 Date-only columns

When source timestamps included time, date-only columns were created with `Date.From([InDate])` and `Date.From([OutDate])` to support relationships with `DimDate[Date]`.

7.3.2 Current open records

A record is provisionally current when an entry date exists and the out date is blank. This is valid only if the export is current and blank out date reliably means the unit is still inside.

7.3.3 ActiveAgeDays

```
if [InDateOnly] = null or [OutDateOnly] <> null then
    null
else
    Duration.Days(Date.From(DateTime.LocalNow()) - [InDateOnly])
```

A fixed extract-date parameter is recommended for reproducibility in production.

7.3.4 ActiveAgeBucket

Open records were grouped into 0–3, 4–7, 8–14, 15–30, and 31+ day buckets. A numeric sort column enforces business order.

7.4 QCWorkingHoursSource handling

The source QC hours were kept but not summed. The column was set to Do not summarize and hidden from report authors until the separate QC fact was created.

Semantic Model and Relationships

8.1 Star schema

The final first-model semantic layer consists of one central container-operations fact, dimensions, a separate QC-performance fact, and a dedicated date table. Dimensions filter facts in a single direction. There are no direct fact-to-fact relationships.

Figure Placeholder

Final Power BI relationship diagram showing FactContainerOperations, FactQCPerformance, dimensions, and active/inactive date relationships

Figure 8.1: Final Power BI relationship diagram showing FactContainerOperations, FactQCPerformance, dimensions, and active/inactive date relationships

8.2 Relationships

Fact endpoint	Dimension endpoint	Cardinality	Status
FactContainerOperations.VesselCallKey	DimVesselCall.VesselCallKey	*:1	Active, single
FactContainerOperations.OperatorKey	DimOperator.OperatorKey	*:1	Active, single
FactContainerOperations.ContainerSpecificationKey	DimContainerSpecification.ContainerSpecificationKey	*:1	Active, single
FactContainerOperations.CargoKey	DimCargo.CargoKey	*:1	Active, single
FactContainerOperations.EquipmentKey	DimEquipment.EquipmentKey	*:1	Active, single
FactContainerOperations.POLKey	DimPortPOL.POLKey	*:1	Active, single
FactContainerOperations.PODKey	DimPortPOD.PODKey	*:1	Active, single
FactContainerOperations.TransportProfileKey	DimTransportProfile.TransportProfileKey	*:1	Active, single
FactContainerOperations.DamageConditionKey	DimDamageCondition.DamageConditionKey	*:1	Active, single
FactContainerOperations.InDateOnly	DimDate.Date	*:1	Active, single
FactContainerOperations.OutDateOnly	DimDate.Date	*:1	Inactive, single
FactQCPerformance.VesselCallKey	DimVesselCall.VesselCallKey	*:1	Active, single
FactQCPerformance.EquipmentKey	DimEquipment.EquipmentKey	*:1	Active, single

8.3 Removed accidental relationships

Auto-detected relationships between PortMaster, DimPortPOL, and DimPortPOD were deleted. Role-playing port dimensions must not filter one another, and a Power Query enrichment table does not need a semantic-model relationship after its attributes have been merged.

8.4 Filter direction rationale

Single-direction dimension-to-fact filtering reduces ambiguity and follows Power BI relationship guidance (Microsoft 2025c). Bidirectional filtering is reserved for an explicit, tested business requirement.

Date Dimension and Time Intelligence

9.1 DimDate construction

`DimDate` was created as a DAX calculated table covering complete calendar years from the earliest entry date to the latest available entry or exit date.

Listing 9.1: Dynamic date table

```
DimDate =  
VAR MinimumInDate =  
    MINX ( FactContainerOperations,  
        FactContainerOperations[InDateOnly] )  
VAR MaximumInDate =  
    MAXX ( FactContainerOperations,  
        FactContainerOperations[InDateOnly] )  
VAR MaximumOutDate =  
    MAXX ( FactContainerOperations,  
        FactContainerOperations[OutDateOnly] )  
VAR MinimumModelDate = DATE ( YEAR ( MinimumInDate ), 1, 1 )  
VAR MaximumAvailableDate = MAX ( MaximumInDate, MaximumOutDate )  
VAR MaximumModelDate = DATE ( YEAR ( MaximumAvailableDate ), 12, 31 )  
RETURN CALENDAR ( MinimumModelDate, MaximumModelDate )
```

Columns include Year, Quarter, YearQuarter, MonthNumber, MonthName, MonthShort, YearMonthNumber, YearMonth, MonthYear, WeekNumber, DayOfMonth, DayOfWeekNumber, DayOfWeek, DayOfWeekShort, IsWeekend, and MonthStartDate. Month and weekday labels are sorted by numeric columns. The table was marked as the official date table, and Auto date/time was disabled (Microsoft 2025d; Microsoft 2025a).

9.2 Role-playing relationships

The entry relationship is active. The exit relationship is inactive and is activated inside exit and completed-dwell measures with `USERELATIONSHIP`. The function requires the exact endpoint columns of an existing relationship (Microsoft 2025e).

9.3 Lesson learned

Early dwell measures failed when they filtered an `IsCompleted` column that was unnecessary and problematic. The final average and median measures rely on blank-aware numeric aggregation plus the inactive exit-date relationship. This is simpler and robust.

Measure Engineering and Model Validation

10.1 Dedicated measure table

A one-row `_Measures` table was created. At least one DAX measure was moved into it before the placeholder column was hidden; otherwise, hiding the table's only visible column makes the table disappear from Report view. Measures were organized into display folders.

Figure Placeholder

The `_Measures` table organized into Volume, Flow, Inventory, Dwell, Data Quality, Equipment and QC, Prior Period, and Technical folders

Figure 10.1: The `_Measures` table organized into Volume, Flow, Inventory, Dwell, Data Quality, Equipment and QC, Prior Period, and Technical folders

10.2 Technical validation measures

```
Source Records = COUNTROWS ( FactContainerOperations )
```

```
Unique Operation Keys =  
DISTINCTCOUNT ( FactContainerOperations[ContainerOperationKey] )
```

```
Duplicate Operation Rows =  
[Source Records] - [Unique Operation Keys]
```

```
Blank Operation Keys =  
COALESCE (
```

```

CALCULATE (
    COUNTROWS ( FactContainerOperations ),
    ISBLANK ( FactContainerOperations[ContainerOperationKey] )
),
0
)

```

The validation page showed 98,052 source records, 98,052 unique operation keys, zero duplicate operation keys, and zero blank keys.

10.3 Core flow measures

```

Unique Containers =
DISTINCTCOUNT ( FactContainerOperations[ContainerNumber] )

```

```

Container Entries =
CALCULATE (
    COUNTROWS ( FactContainerOperations ),
    NOT ISBLANK ( FactContainerOperations[InDateOnly] )
)

```

```

Container Exits =
CALCULATE (
    COUNTROWS ( FactContainerOperations ),
    USERELATIONSHIP (
        DimDate[Date],
        FactContainerOperations[OutDateOnly]
    ),
    NOT ISBLANK ( FactContainerOperations[OutDateOnly] )
)

```

```

Net Container Flow = [Container Entries] - [Container Exits]

```

10.4 Inventory measures

```

Current Records Inside =
CALCULATE (
    COUNTROWS ( FactContainerOperations ),
    REMOVEFILTERS ( DimDate ),
    NOT ISBLANK ( FactContainerOperations[InDateOnly] ),
    ISBLANK ( FactContainerOperations[OutDateOnly] )
)

```

```

Current Unique Containers Inside =
CALCULATE (
    DISTINCTCOUNT ( FactContainerOperations[ContainerNumber] ),
    REMOVEFILTERS ( DimDate ),
    NOT ISBLANK ( FactContainerOperations[InDateOnly] ),
    ISBLANK ( FactContainerOperations[OutDateOnly] )
)

```

```
)
```

These measures intentionally remain unchanged when the date slicer changes because they represent the latest extract state.

10.5 Completed dwell measures

```
Completed Records =  
CALCULATE (  
    COUNTROWS ( FactContainerOperations ),  
    USERRELATIONSHIP (  
        DimDate[Date],  
        FactContainerOperations[OutDateOnly]  
    ),  
    NOT ISBLANK ( FactContainerOperations[CompletedDwellDays] )  
)  
  
Average Completed Dwell Days =  
CALCULATE (  
    AVERAGE ( FactContainerOperations[CompletedDwellDays] ),  
    USERRELATIONSHIP (  
        DimDate[Date],  
        FactContainerOperations[OutDateOnly]  
    )  
)  
  
Median Completed Dwell Days =  
CALCULATE (  
    MEDIAN ( FactContainerOperations[CompletedDwellDays] ),  
    USERRELATIONSHIP (  
        DimDate[Date],  
        FactContainerOperations[OutDateOnly]  
    )  
)
```

The measures are evaluated by exit date and exclude incomplete records because blank dwell values are ignored.

11

TEU, Weight, and QC Validation

11.1 TEU mapping

Container length was inspected in `DimContainerSpecification`. The approved mapping implemented in the project is:

ContainerLength	TEUFactor
20	1.00
40	2.00
45	2.25
UNKNOWN	blank

The old null-only TEUFactor column was removed, the new custom column was renamed TEUFactor, and TEUMappingStatus was recreated after the final factor step.

```
let
    LengthClean =
        if [ContainerLength] = null then null
        else Text.Upper(Text.Trim(Text.Clean(Text.From([ContainerLength]))))
in
    if LengthClean = "20" then 1
    else if LengthClean = "40" then 2
    else if LengthClean = "45" then 2.25
    else null
```

11.2 TEU measures

```
Total TEU =
SUMX (
    FactContainerOperations,
    RELATED ( DimContainerSpecification[TEUFactor] )
)

TEU Entries =
CALCULATE (
    [Total TEU],
    NOT ISBLANK ( FactContainerOperations[InDateOnly] )
```

```

)

TEU Exits =
CALCULATE (
    [Total TEU],
    USERRELATIONSHIP ( DimDate[Date],
        FactContainerOperations[OutDateOnly] ),
    NOT ISBLANK ( FactContainerOperations[OutDateOnly] )
)

Net TEU Flow = [TEU Entries] - [TEU Exits]

Current TEU Inside =
CALCULATE (
    [Total TEU],
    REMOVEFILTERS ( DimDate ),
    NOT ISBLANK ( FactContainerOperations[InDateOnly] ),
    ISBLANK ( FactContainerOperations[OutDateOnly] )
)

```

Figure Placeholder

TEU Validation page showing total TEU, entry/exit TEU, current TEU, mapping completeness, and specification-level reconciliation

Figure 11.1: TEU Validation page showing total TEU, entry/exit TEU, current TEU, mapping completeness, and specification-level reconciliation

11.3 Weight validation

WeightValue is treated as a source-unit measure until the terminal confirms whether it is kilograms, tonnes, gross weight, cargo weight, or verified gross mass. WeightDataStatus classifies missing, negative, zero, and valid positive values. The dashboard labels all weight measures as source unit.

11.4 QC working-hour grain validation

The source was grouped by vessel-call attributes and EquipmentNumber. Each group received row count, minimum, maximum, nested AllRows, distinct nonblank QC-hour count, blank count, and a status. The project found 253 groups with multiple QC working-hour values. All remaining groups contained one consistent repeated value, and no groups were missing QC hours.

11.5 FactQCPerformance

A second fact was created at one row per vessel call and equipment. Consistent groups use the single repeated value as `ValidatedQCWorkingHours`. Inconsistent groups retain blank validated hours and an invalid flag. Handled source records, unique containers, and TEU were aggregated from nested rows. The fact connects to `DimVesselCall` and `DimEquipment`, not directly to `FactContainerOperations`.

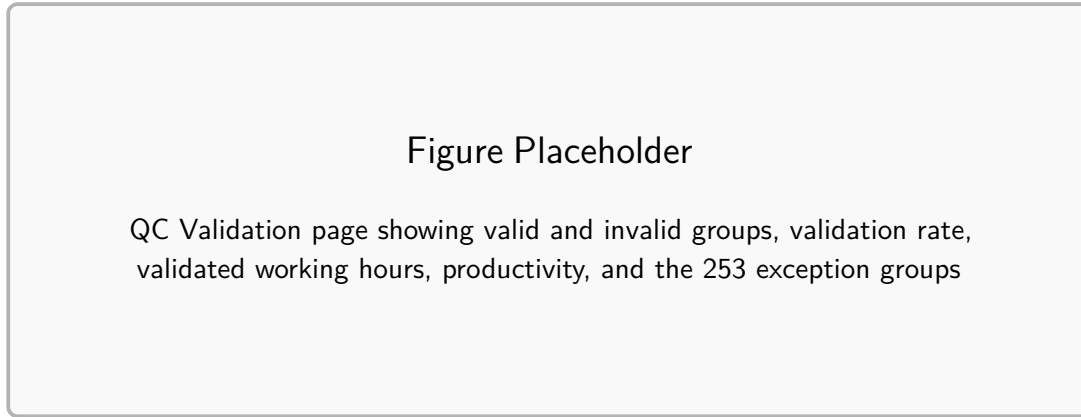


Figure 11.2: QC Validation page showing valid and invalid groups, validation rate, validated working hours, productivity, and the 253 exception groups

Container Flow and Dwell Performance Dashboard

12.1 Business objective

The page combines historical movement, completed dwell, current open inventory, active aging, operator and cargo segmentation, and long-stay detail. Historical visuals use DimDate; current-state visuals deliberately ignore or disable the date slicer.

12.2 Page components

- KPI cards: entries, exits, net flow, current open records, TEU entries/exits, average and median dwell.
- Monthly entry/exit and net-flow combination chart.
- Monthly TEU entry and exit trend.
- Average, median, and P90 dwell by exit month.
- Operator dwell ranking with a minimum completed-record threshold.
- Cargo and commodity matrix with conditional formatting.
- Current open-record aging buckets.
- Longest-aging open-container table and container drillthrough.

Figure Placeholder

Container Flow and Dwell Performance dashboard with historical and latest-state sections

Figure 12.1: Container Flow and Dwell Performance dashboard with historical and latest-state sections

12.3 Design decisions

The page title states the operational purpose. Date interactions are disabled for current inventory and aging visuals. Dwell charts explicitly state *by exit month*. Average and median are shown together because long-stay outliers can pull the average above the typical experience. P90 is included to expose the slower tail without relying on one maximum value.

12.4 Limitations

Current inventory is inferred from blank OutDate and is not a historical snapshot. Active age based on local refresh time should eventually use a fixed extract timestamp. Dwell thresholds are not classified as good or bad until official terminal service targets are supplied.

Executive Overview Dashboard

13.1 Executive objective

The Executive Overview summarizes volume, TEU, current inventory, dwell, vessel activity, QC analytical readiness, operator and cargo concentration, and data-quality exceptions. It avoids operational detail while retaining drill paths to validation and analysis pages.

13.2 Layout

The page uses a 16:9 canvas, a restrained theme, title and subtitle, five high-value slicers, two KPI rows, monthly flow and dwell charts, top operator and vessel workload visuals, cargo mix, inventory aging, and a compact quality section.

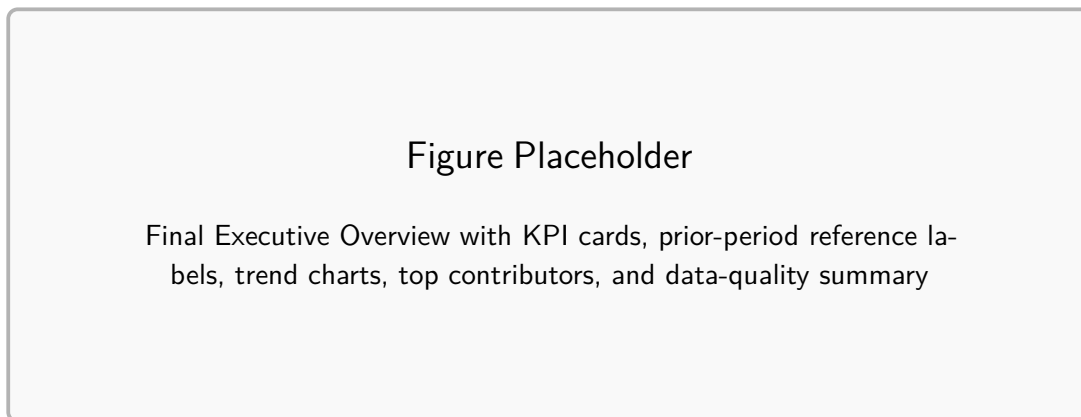


Figure 13.1: Final Executive Overview with KPI cards, prior-period reference labels, trend charts, top contributors, and data-quality summary

13.3 Prior-period indicators

Compact reference labels were added to selected cards rather than creating separate large visuals. Comparisons include entries, TEU entered, and average completed dwell. Historical inventory comparison was not created because no snapshot fact exists.

13.3.1 Latest-year TEU comparison

A simple `TEU Entered Current Year = [TEU Entries]` was rejected because it returns all visible years when no DimDate year filter is applied. The final pattern explicitly determines the latest operational year, clears only date filters, and applies that year.

```

Latest Operational Year =
CALCULATE (
    MAX ( DimDate[Year] ),
    FILTER (
        ALL ( DimDate ),
        CALCULATE ( [TEU Entries] ) > 0
    )
)

TEU Entered Current Year =
VAR CurrentOperationalYear = [Latest Operational Year]
RETURN
    CALCULATE (
        [TEU Entries],
        REMOVEFILTERS ( DimDate ),
        DimDate[Year] = CurrentOperationalYear
    )

TEU Entered Prior Year =
VAR CurrentOperationalYear = [Latest Operational Year]
RETURN
    CALCULATE (
        [TEU Entries],
        REMOVEFILTERS ( DimDate ),
        DimDate[Year] = CurrentOperationalYear - 1
    )

TEU Entered Change % =
DIVIDE (
    [TEU Entered Current Year] - [TEU Entered Prior Year],
    [TEU Entered Prior Year]
)

```

Only DimDate filters are removed; operator, cargo, container length, ports, and other business slicers continue to apply.

13.4 Card implementation

The current Card visual is used with a main callout value and reference labels. Reference measures are renamed for the visual as *Prior Year* and *Change*. Dwell uses reversed directional interpretation: a negative variance is generally favorable, while higher volume is treated as directional rather than automatically good.

Data Quality, Governance, and Auditability

14.1 Quality principles

The solution preserves unknowns, separates technical validation from business validation, and avoids replacing missing facts with zero. Match-status columns monitor every dimension merge. Validation measures report technical-key uniqueness, TEU mapping completeness, missing weight, and QC consistency.

14.2 Primary rule categories

- Identity: missing container number, exact duplicate, conflicting candidate visit.
- Date: missing entry, out before in, future dates, long duration, stack-days variance.
- Vessel call: call key mapping to multiple vessel or voyage descriptions.
- Container specification: missing length, inconsistent size/type, unmapped TEU.
- Weight: missing, zero, negative, unit inconsistency, extreme values.
- Port: unknown code, missing master mapping, invalid name-code pairing.
- QC: negative or zero hours, repeated values, multiple values within a call/equipment group.

14.3 Security and privacy

Future production deployment should classify container identifiers, customer details, dangerous-goods information, claims, and commercial data. RLS may restrict external operators or customers; OLS may protect sensitive financial or safety columns. Workspace roles must be designed carefully because RLS behavior depends on service roles.

14.4 Audit trail

Every mapping table should record owner, effective date, source, and approval. Power Query Applied Steps provide technical lineage; this book provides business and implementation rationale. Produc-

tion releases should include row-count reconciliation, KPI regression testing, relationship validation, refresh duration, and sign-off.

Performance, Deployment, and Maintenance

15.1 Model optimization

The current model is small enough for Import mode and full refresh. Optimization practices include disabling load for staging and reference-only queries, removing unused columns, using integer keys, avoiding bidirectional relationships, hiding technical columns, using explicit measures, and preventing high-cardinality text from proliferating in facts.

15.2 Deployment lifecycle

The recommended lifecycle is Development, Test/UAT, and Production. A deployment pipeline or equivalent release process should parameterize source paths, test refresh and security, compare KPI baselines, and maintain release notes.

15.3 Fabric migration path

A future Microsoft Fabric implementation can move persistent surrogate-key management, slowly changing dimensions, large history, snapshots, and transformation logic to a Lakehouse or Warehouse. Power BI then becomes a thin semantic and presentation layer over curated data products.

15.4 Maintenance procedure

1. Update the TOS export and governed Excel masters.
2. Refresh Power Query previews and resolve missing mappings.
3. Run model, TEU, weight, and QC validation pages.
4. Compare row counts and key KPIs with the previous release.
5. Review new codes, unknowns, and inconsistent QC groups.
6. Publish only after validation acceptance.

Conclusions and Future Roadmap

16.1 Achievements

The project converted a wide operational export into a validated enterprise semantic model, introduced reusable dimensions, established date-role intelligence, prevented QC-hour overstatement, mapped TEU including 45-foot containers at 2.25, created a structured measure library, and built validation, operational, and executive reporting pages.

16.2 Key lessons learned

- Data volume was not the main challenge; grain and repeated values were.
- Technical uniqueness and business uniqueness are different tests.
- USERRELATIONSHIP must reference the exact existing relationship columns.
- Blank-aware averages and medians were more reliable than an unnecessary IsCompleted filter.
- Current inventory cannot be compared historically without snapshots.
- Validation pages should precede executive presentation.

16.3 Future enhancements

Priority enhancements include a periodic inventory snapshot, detailed yard locations and capacities, vessel planned and actual timestamps, berth intervals, gate and truck milestones, train-call facts, equipment state intervals, inspection and damage event facts, customer dimensions, tariffs and financial facts, approved DG rules, and persistent warehouse-managed surrogate keys.

Project Placeholder

Complete the project-specific ownership matrix, production source locations, refresh schedule, approved KPI targets, security roles, support contacts, and release process before production handover.

Complete DAX Measure Library

This appendix consolidates the final governed measures. Measures that depend on unavailable business definitions are marked as future.

A.1 Technical and volume

```
Source Records = COUNTROWS ( FactContainerOperations )
Unique Operation Keys = DISTINCTCOUNT ( FactContainerOperations[ContainerOperationKey] )
Duplicate Operation Rows = [Source Records] - [Unique Operation Keys]
Blank Operation Keys = COALESCE ( CALCULATE ( COUNTROWS ( FactContainerOperations
    ), ISBLANK ( FactContainerOperations[ContainerOperationKey] ) ), 0 )
Unique Containers = DISTINCTCOUNT ( FactContainerOperations[ContainerNumber] )
Vessel Calls = DISTINCTCOUNT ( FactContainerOperations[VesselCallKey] )
```

A.2 Flow and inventory

```
Container Entries = CALCULATE ( COUNTROWS ( FactContainerOperations ), NOT ISBLANK
    ( FactContainerOperations[InDateOnly] ) )
Container Exits = CALCULATE ( COUNTROWS ( FactContainerOperations ),
    USERELATIONSHIP ( DimDate[Date], FactContainerOperations[OutDateOnly] ), NOT
    ISBLANK ( FactContainerOperations[OutDateOnly] ) )
Net Container Flow = [Container Entries] - [Container Exits]
Current Records Inside = CALCULATE ( COUNTROWS ( FactContainerOperations ),
    REMOVEFILTERS ( DimDate ), NOT ISBLANK ( FactContainerOperations[InDateOnly] ),
    ISBLANK ( FactContainerOperations[OutDateOnly] ) )
Current Unique Containers Inside = CALCULATE ( DISTINCTCOUNT ( FactContainerOperations
    [ContainerNumber] ), REMOVEFILTERS ( DimDate ), NOT ISBLANK (
    FactContainerOperations[InDateOnly] ), ISBLANK ( FactContainerOperations[OutDateOnly] )
    )
```

A.3 Dwell

Completed Records = **CALCULATE** (**COUNTROWS** (FactContainerOperations),
USERELATIONSHIP (DimDate[Date], FactContainerOperations[OutDateOnly]), **NOT**
ISBLANK (FactContainerOperations[CompletedDwellDays]))

Average Completed Dwell Days = **CALCULATE** (**AVERAGE** (FactContainerOperations[
CompletedDwellDays]), **USERELATIONSHIP** (DimDate[Date], FactContainerOperations[
OutDateOnly]))

Median Completed Dwell Days = **CALCULATE** (**MEDIAN** (FactContainerOperations[
CompletedDwellDays]), **USERELATIONSHIP** (DimDate[Date], FactContainerOperations[
OutDateOnly]))

Maximum Completed Dwell Days = **CALCULATE** (**MAX** (FactContainerOperations[
CompletedDwellDays]), **USERELATIONSHIP** (DimDate[Date], FactContainerOperations[
OutDateOnly]))

P90 Completed Dwell Days = **CALCULATE** (**PERCENTILEX.INC** (**FILTER** (FactContainerOperations, **NOT** **ISBLANK** (FactContainerOperations[CompletedDwellDays])), FactContainerOperations[CompletedDwellDays], 0.90), **USERELATIONSHIP** (DimDate[Date], FactContainerOperations[OutDateOnly]))

A.4 TEU

Total TEU = **SUMX** (FactContainerOperations, **RELATED** (DimContainerSpecification[TEUFactor]))

TEU Entries = **CALCULATE** ([Total TEU], **NOT** **ISBLANK** (FactContainerOperations[InDateOnly]))

TEU Exits = **CALCULATE** ([Total TEU], **USERELATIONSHIP** (DimDate[Date], FactContainerOperations[OutDateOnly]), **NOT** **ISBLANK** (FactContainerOperations[OutDateOnly]))

Net TEU Flow = [TEU Entries] - [TEU Exits]

Current TEU Inside = **CALCULATE** ([Total TEU], **REMOVEFILTERS** (DimDate), **NOT** **ISBLANK** (FactContainerOperations[InDateOnly]), **ISBLANK** (FactContainerOperations[OutDateOnly]))

Records Missing TEU Mapping = **COUNTROWS** (**FILTER** (FactContainerOperations, **ISBLANK** (**RELATED** (DimContainerSpecification[TEUFactor]))))

Records with TEU Mapping = [Source Records] - [Records Missing TEU Mapping]

TEU Mapping Completion % = **DIVIDE** ([Records with TEU Mapping], [Source Records])

A.5 Weight

Total Weight - Source Unit = **CALCULATE** (**SUM** (FactContainerOperations[WeightValue]), FactContainerOperations[WeightValue] > 0)

Average Weight - Source Unit = **CALCULATE** (**AVERAGE** (FactContainerOperations[WeightValue]), FactContainerOperations[WeightValue] > 0)

Median Weight - Source Unit = **CALCULATE** (**MEDIAN** (FactContainerOperations[WeightValue]), FactContainerOperations[WeightValue] > 0)

Records Missing Weight = **CALCULATE** ([Source Records], **ISBLANK** (FactContainerOperations[WeightValue]))

Records with Valid Weight = **CALCULATE** ([Source Records], FactContainerOperations[WeightValue] > 0)

Weight Data Completion % = **DIVIDE** ([Records with Valid Weight], [Source Records])

A.6 QC

```

QC Performance Groups = COUNTROWS ( FactQCPerformance )
Valid QC Groups = CALCULATE ( [QC Performance Groups], FactQCPerformance[
    IsQCWorkingHoursValid] = TRUE () )
Invalid QC Groups = CALCULATE ( [QC Performance Groups], FactQCPerformance[
    IsQCWorkingHoursValid] = FALSE () )
QC Working Hours Validation % = DIVIDE ( [Valid QC Groups], [QC Performance Groups] )
Validated QC Working Hours = SUM ( FactQCPerformance[ValidatedQCWorkingHours] )
QC Handled Source Records = SUM ( FactQCPerformance[ContainerRowCount] )
QC Handled TEU = SUM ( FactQCPerformance[HandledTEU] )
Valid QC Handled Records = CALCULATE ( [QC Handled Source Records], FactQCPerformance[
    IsQCWorkingHoursValid] = TRUE () )
Valid QC Handled TEU = CALCULATE ( [QC Handled TEU], FactQCPerformance[
    IsQCWorkingHoursValid] = TRUE () )
Container Records per QC Working Hour = DIVIDE ( [Valid QC Handled Records], [Validated QC
    Working Hours] )
TEU per QC Working Hour = DIVIDE ( [Valid QC Handled TEU], [Validated QC Working Hours] )

```

A.7 Latest-year comparisons

```

Latest Operational Year = CALCULATE ( MAX ( DimDate[Year] ), FILTER ( ALL ( DimDate ),
    CALCULATE ( [TEU Entries] ) > 0 ) )
TEU Entered Current Year = VAR Y = [Latest Operational Year] RETURN CALCULATE ( [TEU
    Entries], REMOVEFILTERS ( DimDate ), DimDate[Year] = Y )
TEU Entered Prior Year = VAR Y = [Latest Operational Year] RETURN CALCULATE ( [TEU
    Entries], REMOVEFILTERS ( DimDate ), DimDate[Year] = Y - 1 )
TEU Entered Change = [TEU Entered Current Year] - [TEU Entered Prior Year]
TEU Entered Change % = DIVIDE ( [TEU Entered Change], [TEU Entered Prior Year] )

```

B

Power Query M Formula Library

B.1 Common cleaning pattern

```
if [SourceColumn] = null or Text.Trim(Text.From([SourceColumn])) = "" then
    "UNKNOWN"
else
    Text.Upper(Text.Trim(Text.Clean(Text.From([SourceColumn]))))
```

B.2 Completed dwell

```
if [InDate] = null or [OutDate] = null then null
else if [OutDate] < [InDate] then null
else Duration.TotalDays([OutDate] - [InDate])
```

B.3 Date-only columns

```
if [InDate] = null then null else Date.From([InDate])
if [OutDate] = null then null else Date.From([OutDate])
```

B.4 TEU mapping

```
let
    LengthClean = if [ContainerLength] = null then null
                  else Text.Upper(Text.Trim(Text.Clean(Text.From([ContainerLength]))))
in
    if LengthClean = "20" then 1
    else if LengthClean = "40" then 2
    else if LengthClean = "45" then 2.25
    else null
```

B.5 QC distinct-hour count

```
List.Count(  
    List.Distinct(  
        List.RemoveNulls(  
            Table.Column([AllRows], "QCWorkingHoursSource")  
        )  
    )  
)
```

B.6 Validated QC hours

```
if [DistinctQCWorkingHourCount] = 1 then  
    [MaximumQCWorkingHours]  
else  
    null
```

B.7 Active age and buckets

```
if [InDateOnly] = null or [OutDateOnly] <> null then null  
else Duration.Days(Date.From(DateTime.LocalNow()) - [InDateOnly])
```

```
if [ActiveAgeDays] = null then "Not Active"  
else if [ActiveAgeDays] <= 3 then "0-3 Days"  
else if [ActiveAgeDays] <= 7 then "4-7 Days"  
else if [ActiveAgeDays] <= 14 then "8-14 Days"  
else if [ActiveAgeDays] <= 30 then "15-30 Days"  
else "31+ Days"
```

Project Placeholder

Export the complete Advanced Editor scripts from each final Power Query query and replace this placeholder before production handover. The present appendix preserves every custom M expression established during the project but not every automatically generated navigation and type-change step.

C

Data Dictionary

Source / Model Field	Type		Business purpose	Quality or implementation note
ContainerNumber	Text		Container identifier and drillthrough field	Not assumed unique across visits
ContainerOperationKey	Whole number		Technical fact-row key	Generated index; validated unique
VesselCallKey	Whole number		Links container fact to vessel call	Generated by merge
OperatorKey	Whole number		Links to operator	OPR role requires confirmation
ContainerSpecificationKey	Whole number		Links size and type	Supports TEU mapping
CargoKey	Whole number		Links cargo type and commodity	Source commodity preserved
EquipmentKey	Whole number		Links equipment	QC hours excluded from dimension
POLKey / POD-Key	Whole number		Role-playing port keys	Enriched from PortMaster
TransportProfileKey	Whole number		Links status and transport classifications	Junk-dimension combination
DamageConditionKey	Whole number		Links source damage condition	Cause and liability unavailable
InDate / OutDate	Date/Time Date	or	Lifecycle timestamps	Out may be blank for open records
InDateOnly / OutDateOnly	Date		Calendar relationship columns	Entry active, exit inactive
CompletedDwellDays	Decimal		Valid completed dwell	Blank for incomplete/invalid rows
StackDays	Numeric		TOS-provided stack duration	Reconcile with calculated dwell

WeightValue	Decimal	Source weight	Unit not yet confirmed
WeightGroup	Text	Source weight band	Remains in fact
IMDG	Text	Source DG value	Remains in fact; mapping deferred
QCWorkingHoursSource	Decimal	Source QC hours	Never summed in container fact
TEUFactor	Decimal	TEU conversion	20=1, 40=2, 45=2.25
ActiveAgeDays	Whole number	Age of current open record	Should use extract timestamp in production

D

Relationship Documentation and KPI Catalog

D.1 Relationship controls

All relationships are many-to-one, single direction, and active except the exit-date relationship. Port master and role-playing port dimensions have no direct model relationships.

D.2 KPI catalog

KPI	Business objective	Principal limitations
Container Entries	Quantify records entering selected period	Source records not proven physical visits
Container Exits	Quantify records exiting selected period	Uses inactive OutDateOnly relationship
Current Unique Containers Inside	Latest open-container state	Not historical inventory
Total TEU	Capacity-equivalent volume	Unknown lengths excluded
Average/Median/P90 Dwell	Service and congestion performance	Based on completed exits
Vessel Calls	Call workload count	Depends on generated call key
QC Records per Hour	Validated equipment productivity proxy	Not confirmed physical crane moves
TEU per QC Hour	Capacity-oriented QC productivity	Invalid 253 groups excluded
Weight Completion	Weight analytical readiness	Unit and meaning not confirmed
TEU Mapping Completion	Size mapping readiness	Unknown length remains blank

E

Visualization Inventory

Page	Visual	Operational question
00 Model Validation	Technical cards	Does the fact retain the expected rows and unique keys?
01 TEU Validation	KPI cards and specification table	Are TEU factors complete and totals reconcilable?
02 Weight Validation	Weight cards, distribution, matrices	Is weight usable and where is workload concentrated?
03 QC Validation	QC cards, equipment bars, exception table	Are hours reliable and which equipment performs best?
04 Container Flow and Dwell	Flow/TEU trends, dwell trend, aging, detail	How do movement, current inventory, and dwell behave?
05 Executive Overview	KPI cards, prior-period labels, trends, top-N	What requires executive attention now?
DT Container Detail	Detail layout	What is known about the selected container record?

E.1 Figure insertion checklist

Replace the placeholders with approved screenshots after removing or masking sensitive container numbers, customer names, and operational details. Use consistent 16:9 crops and ensure figure text remains legible in print.

F

Glossary of Business and Technical Terms

Active age

Elapsed days from entry to the latest extract for a record without an out date.

Calling year

Year assigned to a vessel call; not necessarily the same as entry or exit year.

Completed dwell

Time between validated entry and exit dates.

Dimension

Descriptive table used to filter and group facts.

Fact Table at a declared measurement grain containing foreign keys and numeric observations.

Grain

Exact business meaning of one row in a table.

Junk dimension

Dimension combining several low-cardinality classifications.

Left outer join

Merge that preserves all rows from the first query and adds matches from the second.

Role-playing dimension

One conformed dimension used in multiple roles, such as POL and POD or entry and exit date.

Surrogate key

Generated technical key independent of a source business code.

TEU

Twenty-foot equivalent unit; project mapping is 20=1, 40=2, and 45=2.25.

Validation page

Technical report page used to confirm quality and reconciliation before business publication.

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